SMART DRIVER & VEHICLE PROACTIVE SAFETY SYSTEM

INDIVIDUAL THESIS

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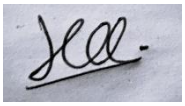
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Abstract

A small closing gap to the car ahead and driver fatigue both increase the probability of a crash. However, driver status monitoring and forward collision warning are often developed as independent services. The thesis proposes the Smart Driver & Vehicle Proactive Safety System, an Internet of Things (IoT) unit for tourism transport in Sri Lanka that evaluates the two threats combined. An ESP32-CAM runs MobileNetV2 eye-state and yawn models on device, an HC-SR04 ultrasonic sensor measures the distance to the vehicle ahead, and an ESP32 main hub receives both over ESP-NOW and fuses them into a Composite Safety Index (CSI) built from the Eye Aspect Ratio (EAR) and the Time-to-Collision (TTC). Alerts grow with risk: an audible warning, a real-time driver app dashboard with a live map, and a Firebase Cloud Messaging (FCM) push notice, comprising only the alert type and timestamp, to the fleet administrator. Only at the highest risk level is steering wheel vibration utilized to reduce attention. No video or image leaves the camera, and no independent inference server is employed; instead, the hub transmits data to the Firebase Realtime Database. Cellular connectivity and the vehicle position for the map are provided by a SIM7600CE module. The yawn model (YawDD) had a best validation accuracy of 99.34% and a test accuracy of 99.39% on public datasets, while the eye-state model (MRL Eye) achieved a best validation and test accuracy of 99.21%. These are solely offline results. The entire system has not undergone bench, in-car, or user validation. The thesis documents the requirements, design, test plan, limits and commercial prospects, and highlights validation on the deployed hardware as the key future work.

Keywords: driver fatigue detection, time-to-collision, sensor fusion, MobileNetV2, Internet of Things

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List of Abbreviations

Abbreviation	Full Form
AI	Artificial intelligence
CCMP	Counter Mode with CBC-MAC Protocol
CSI	Composite Safety Index
EAR	Eye Aspect Ratio
FCM	Firebase Cloud Messaging
FR	Functional requirement
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
IoT	Internet of Things
JSON	JavaScript Object Notation
LTE	Long-Term Evolution
NFR	Non-functional requirement
PERCLOS	Percentage of eyelid closure over time
RQ	Research question
SDG	Sustainable Development Goal
SLIIT	Sri Lanka Institute of Information Technology
TC	Test case
TTC	Time-to-Collision
WBS	Work breakdown structure
WHO	World Health Organization
YawDD	Yawning Detection Dataset

1 INTRODUCTION

1.1 Overall Project Context

This thesis details my personal contribution to the group research project "A Next-Generation AI Framework for Intelligent Travel Support in Sri Lanka", which was conducted by group R26-IT-112 at the Sri Lanka Institute of Information Technology (SLIIT). In order to consider tourism services that are often handled individually, the group framework combines itinerary planning, real-time response to disturbances, tourism-flow forecasts, in-transit driver safety, and compliance assistance. One component was created by each group member. The results of the other components belong to their respective authors and are not presented as mine; only the in-transit driver safety component is reported here.

Road transport is crucial to tourism in Sri Lanka, and tourists often travel long distances on narrow rural or mountain roads, where driver weariness shortens reaction time and escalates the danger of a crash. The World Health Organization (WHO) estimates that 1.19 million people die in automobile accidents annually [1] , indicating the scope of the issue. One of the Sustainable Development Goals of the United Nations is to cut the number of road traffic deaths and injuries worldwide in half [2]. According to the WHO, this goal is currently set for 2030 [1]. A safety unit that helps a tourist-vehicle driver stay alert, and helps the operator know when a vehicle is at risk, addresses a real need in the setting of this project.

1.2 Individual Component Context

My component is the Smart Driver & Vehicle Proactive Safety System. It is an Internet of Things (IoT) edge unit installed in the vehicle. It has three main parts. **First**, a single ESP32-CAM board with an OV2640 camera captures the driver's face and runs two MobileNetV2 models, an eye-state model (open or closed) and a yawn model (yawn or no yawn), on the device itself. The camera outputs only a status, not images. **Second**, an HC-SR04 ultrasonic sensor measures the distance to the vehicle ahead. **Third**, an ESP32 main hub receives the eye and yawn status and the distance readings over ESP-NOW, a connectionless wireless protocol from Espressif [3], and fuses them into a single Composite Safety Index (CSI). A SIM7600CE module provides 4G connectivity and the Global Positioning System (GPS) position of the vehicle [4].

The hub sends the data to Firebase Realtime Database [5] , and the driver's mobile app receives it in real time. Alerts are tiered by risk. The system issues in-cabin alerts, updates the driver's app dashboard, and sends a Firebase Cloud Messaging (FCM) push notification [6] to the fleet administrator. The push notification carries only the alert type and a timestamp. Steering wheel vibration is used only at the highest risk level. No video or image stream is sent to any server, and no separate artificial intelligence (AI) inference server is used. The system architecture is described in Chapter 4 and shown in Figure 4.2.

The group paper [1] uses some terms that I read differently in this thesis. Its architecture figure shows "In-Cabin Audio Alarms (Driver Alerts)" and "Cloud Telematics Dashboard (Fleet Safety Telemetry)". In this thesis the first box means in-cabin alerts together with the driver app dashboard, and the second means the push notification sent to the fleet administrator. Where the group paper refers to a fleet-management portal or to remote alerts, I write "the fleet administrator receiving the FCM push notification".

1.3 Problem Statement

Two hazards are especially relevant to a tourist-vehicle driver on a long journey. The first is the driver's own state: tiredness shows in the eyes and in yawning. The second is the traffic condition ahead, where a short or rapidly decreasing distance from the vehicle in front can significantly reduce the driver's available reaction time. Existing research commonly addresses these two risks independently, with driver-state monitoring treated as one area of study [7], [8], [9] and forward collision avoidance addressed separately [10]. However, fatigue does not always result in the same level of risk; a mildly fatigued driver may remain relatively safe on an open road, but the same condition can become hazardous when following another vehicle at a short distance. Therefore, monitoring either driver fatigue or collision risk alone may not provide a complete representation of the vehicle's overall safety condition.

In addition to these safety challenges, practical considerations must also be addressed. Continuous camera-based driver monitoring introduces privacy concerns, making it important to ensure that video data remains within the vehicle. Safety warnings must be delivered promptly while avoiding unnecessary distraction, and excessive false alarms may cause drivers to disregard future warnings. Although fleet administrators need to be informed when a vehicle enters a high-risk condition, they do not necessarily require access to the driver's facial images or, within the proposed design, the vehicle's location through the notification.

Therefore, the problem addressed by this thesis is how to jointly assess driver fatigue and forward collision risk using a low-cost, edge-based system and combine these risks into meaningful safety information. The system should provide proportionate and timely alerts to both the driver and fleet administrator while preserving driver privacy by preventing video data from leaving the vehicle.

1.4 Research Gap

As established in Chapter 2, the existing literature on driver safety follows a distinct pattern. Research into fatigue and drowsiness typically focuses on physical, vehicular, biological, and hybrid indicators [7], [8], [9], [11]. Meanwhile, studies on collision avoidance largely center on traffic assessment and intervention timing, and sensor-fusion research prioritizes detecting objects surrounding the vehicle [12]. Even when commercial fleet systems provide both drowsiness detection and forward collision warnings, they treat them as isolated functions that trigger entirely separate alerts [13].

This thesis, therefore, targets a highly specific gap within this research landscape. Within the scope of the reviewed literature, there is a clear absence of a system that (a) merges a camera-based fatigue metric with a distance-based collision metric into a single, unified risk index, (b) processes this data locally on affordable edge hardware, (c) preserves driver privacy by keeping video data entirely on the device, and (d) connects this index to a tiered warning system where the most disruptive feedback such as steering wheel vibration is strictly reserved for critical danger.

The primary contribution of this work rests in this unique integration and the structured alert mechanism, rather than the invention of new individual technologies. The foundational techniques utilized in this study, such as the Eye Aspect Ratio (EAR) [14], MobileNetV2 , and time-to-collision calculations, are already well-established.

1.5 Aim and Objectives

Aim To design and evaluate an IoT-based driver and vehicle proactive safety unit that combines fatigue detection and forward collision risk into one Composite Safety Index and issues tiered alerts to the driver and the fleet administrator.

To achieve this aim, the following **objectives** have been established:

1. **Model Training and Evaluation:** Train and assess MobileNetV2 models using public datasets to accurately classify eye states and yawning.
2. **Metric Definition:** Establish a fatigue detection metric based on the Eye Aspect Ratio (EAR) and a collision-risk metric using Time-to-Collision (TTC), and subsequently merge them into the unified CSI.
3. **Architecture Design:** Develop a privacy-focused IoT architecture where visual classification models operate locally on an ESP32-CAM. Sensor data will be transmitted to a central ESP32 hub via ESP-NOW, ensuring that no video or image files are ever uploaded to an external server.
4. **Alert System Development:** Create a tiered warning mechanism that intensifies as the calculated risk increases. Physical interventions, such as steering wheel vibrations, will be strictly reserved for critical threats, while fleet administrators will only receive minimal, privacy-conscious notifications containing the alert type and a timestamp.
5. **System Specification and Testing:** Outline comprehensive functional and non-functional requirements, alongside a rigorous test plan designed to validate every stage of the alert system.
6. **Limitation Assessment:** Critically evaluate the proposed design to identify any underlying technical, security, social, and ethical limitations.
7. **Commercial Viability:** Analyze the commercial potential and practical application of implementing this safety system within Sri Lanka's tourism transport sector.

1.6 Research Questions or Hypotheses

The work is engineering-led and is guided by four research questions (RQ) rather than by statistical hypotheses.

1. **RQ1.** Can lightweight MobileNetV2 classifiers reach high accuracy on public eye-state and yawn datasets?
2. **RQ2.** Can EAR-based fatigue and TTC-based collision risk be combined into one index that supports tiered alerts?
3. **RQ3.** Can the alert design limit driver distraction by reserving steering wheel vibration for the highest risk level?
4. **RQ4.** Can the architecture keep the driver's video on the device and still inform the driver and the fleet administrator in real time?

Only RQ1 is answered with measured results in this thesis , and only for offline evaluation on public datasets. RQ2 to RQ4 are answered at the level of design and test planning;

1.7 Scope Limitations and Assumptions

1.7.1 In scope

This research focuses specifically on the development and integration of the safety unit detailed in Section 1.2. The core components encompassed within this project include the ESP32-CAM node (which processes the on-device models), the HC-SR04 ultrasonic sensor for distance measurement, and the central ESP32 hub responsible for calculating the Composite Safety Index (CSI). Additionally, the scope covers the SIM7600CE module for cellular connectivity and GPS tracking, the Firebase Realtime Database, the driver's mobile application featuring a live map, and the Firebase Cloud Messaging (FCM) alert system for fleet administrators.

1.7.2 Out of scope

- Face-based (Face ID) login or any other biometric authentication of the driver.
- Automatic contact with hospitals, the police or any emergency service.
- The other components of the group framework (itinerary optimisation, real-time adaptation, tourism-flow forecasting, emotion recognition, immigration compliance).
- Transmission of video or images to any server, and any separate AI inference server.
- Vehicle control: the system advises the driver and never brakes or steers the vehicle.

1.7.3 Assumptions

- The camera is mounted so that it views the driver’s face, and the ultrasonic sensor faces forward.
- The vehicle supplies power to the units while it is in use.
- Cellular coverage is available for the data link and GPS reception is available for the live map, at least part of the time.
- The driver has the app installed, and the fleet administrator has agreed to receive notifications.
- The public datasets are representative enough for an offline comparison.

1.7.4 Limitations

A primary limitation of this work is that the quantitative results presented are derived entirely from the offline evaluation of the two classification models. The current scope does not include physical bench testing, live in-vehicle trials, interactive user studies, or live on-device performance measurements. To maintain transparency, these constraints are explicitly acknowledged throughout the thesis wherever they might impact the study's conclusions.

1.8 Individual Contribution

The design, code, models and results in this thesis are my own work. My contribution is:

- A targeted literature review and the identification of the research gap (Chapters 1 and 2).
- The design of the system architecture, the CSI, and the tiered alert logic (Chapters 3 and 4).
- The training and evaluation of the MobileNetV2 eye-state and yawn models (Chapters 3 and 6).
- The requirements, the test plan and the description of the implementation (Chapters 4 and 5).
- The analysis of limitations, ethics, privacy and security (Sections 3.6 and 7.5).
- The commercialisation analysis and Sustainable Development Goal (SDG) mapping (Chapter 8).

The other components of the group project were developed by the other members of the group and are outside this thesis.

1.9 Thesis Structure

The remainder of this thesis is structured to seamlessly guide the reader through the research process. Chapter 2 establishes the theoretical foundation by reviewing existing literature and identifying the research gap, while Chapter 3 outlines the methodology, data, and ethical framework. The technical core is presented in Chapters 4 and 5, which detail the system's requirements, architectural design, practical implementation, and testing strategy. Following the development phase, Chapter 6 reports the validation results, leading into a critical discussion of the findings and limitations in Chapter 7. Finally, Chapter 8 analyzes the system's commercial and sustainability potential, before Chapter 9 concludes the study with future research directions, followed by the references, glossary, and appendices.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review Method and Scope

This is a targeted narrative review, not a systematic review. I searched for work on driver fatigue and drowsiness detection, driver monitoring, forward collision avoidance, sensor fusion in vehicles, and IoT fleet safety, and I selected sources that a supervisor or examiner could check. The academic sources were published between 2008 and 2023 in journals such as the IEEE Transactions on Intelligent Transportation Systems, the IEEE Transactions on Vehicular Technology and *Sensors*, or in established conference proceedings. I also used manufacturer datasheets and official documentation for the hardware and cloud services, dataset pages, and one commercial product announcement. Every reference was checked against a publisher, DOI, dataset or manufacturer record before it was included, and works that I could not verify were left out.

The review does not claim to be exhaustive. Its purpose is to establish the concepts the design relies on, to compare closely related work (Section 2.3), and to state the gap (Section 2.5) in a way that the reviewed evidence supports.

2.2 Domain and Theoretical Foundations

2.2.1 Driver fatigue and its visual indicators

Reviews of driver fatigue detection group the methods by the kind of feature they use. Sikander and Anwar describe physical, vehicular, biological, subjective and hybrid features [7]. For a low-cost in-vehicle unit, image-based measures are attractive because they need only a camera and do not touch the driver.

Two visual indicators are used in this work. The first is eye closure. The Eye Aspect Ratio (EAR) of Soukupová and Čech is a single number computed from six landmarks around the eye that falls when the eye closes [10]. The percentage of eyelid closure over time (PERCLOS) is a related measure based on how long the eyes stay closed. The second indicator is yawning, for which the YawDD dataset provides in-car video of drivers [12].

2.2.2 Deep learning on constrained hardware

MobileNetV2 is an efficient convolutional neural network specifically optimized for mobile applications. By utilizing inverted residual blocks and lightweight depth wise convolutions, it significantly reduces computational demands without sacrificing accuracy [11]. Through transfer learning, this pre-trained network can be easily adapted to specific binary classification tasks such as detecting whether an eye is open or if a driver is yawning. Ultimately, this combination of a lightweight architecture and targeted transfer learning makes a MobileNetV2 backbone far better suited for constrained edge devices than heavier, traditional networks.

2.2.3 Forward collision risk

Forward collision avoidance systems continuously monitor the environment ahead of a vehicle to determine exactly when a warning or intervention is necessary. While some researchers, such as Jansson and Gustafsson [8], have proposed complex decision-making frameworks based on nonlinear filtering, a more straightforward and widely adopted metric is Time-to-Collision (TTC). TTC estimates the time remaining before an impact occurs, assuming the current closing speed remains constant. By calculating this metric using the physical gap and relative speed between the vehicles, the system can quickly determine danger levels, where a shorter TTC inherently signals a higher risk.

To gather the distance data required for this TTC calculation, ultrasonic sensors like the HC-SR04 are frequently utilized. This specific sensor measures proximity by timing how long it takes for an emitted ultrasonic pulse to bounce back from an object. According to the manufacturer's datasheet, it operates with an effective range of 2 to 400 cm, a narrow 15-degree measuring angle, and a recommended minimum measurement cycle of 60 ms [13]. As detailed later in Section 7.5, these specific hardware limitations play a critical role in how accurately TTC can be computed in real-time.

2.2.4 Sensor fusion

Sensor fusion combines several sensors so that the result is more reliable than any one of them. The fusion in this thesis is much simpler than fusion of raw sensor data: it combines a driver-state measure and a distance-based measure in a weighted index, that is, fusion at the level of risk indicators.

2.2.5 IoT platform services

To construct the system's communication and data infrastructure, several specialized IoT technologies are integrated. Espressif’s ESP-NOW provides a connectionless Wi-Fi protocol that allows local devices to exchange data directly without relying on an access point while securing these transmissions using CCMP encryption [3]. For cloud management, the Firebase Realtime Database stores system information in JSON format and instantly synchronizes it across all connected clients [5], whereas Firebase Cloud Messaging (FCM) is utilized to reliably push targeted alerts to end-user devices [6]. Finally, wide-area cellular connectivity and location tracking are handled by the SIM7600CE, an LTE Cat-4 module equipped with built-in GNSS capabilities [4].

2.3 Related Research

Table 2.1 compares the closest works. The columns show whether a work addresses driver fatigue, forward collision or both, what sensing and platform it uses, and how it differs from the present work. The table contains only works that I verified. Review papers are included because they define the state of the art for fatigue detection, but I did not use them as evidence for the performance of any specific method.

Table 2.1: Comparison of related work with this thesis

Work	Year	Focus	Fatigue	Collision	Relation to this thesis
Soukupová and Čech [10]	2016	EAR from facial landmarks (blink detection)	Yes (eye cue)	No	Source of the EAR measure used here; no collision component.
Jansson and Gustafsson [8]	2008	Collision avoidance decision making with nonlinear filtering	No	Yes	Forward risk only; no driver-state input.
Sikander and Anwar [7]	2019	Review of driver fatigue detection	Yes (review)	No	Frames the fatigue measures; not a system.
Rudrusamy et al. [14]	2023	IoT fleet safety: iris-based drowsiness, lane tracking, GPS, fleet-centre alerts	Yes	Lane tracking, not forward gap	Closest IoT fleet system; uses lane tracking and face recognition, no TTC or composite index.

Work	Year	Focus	Fatigue	Collision	Relation to this thesis
Commercial fleet dashcam [9]	n/a	AI dashcam with drowsiness detection and forward collision warning	Yes	Yes	Both functions offered, described as separate alerts.
This thesis	2026	CSI from EAR and TTC on ESP32 edge hardware	Yes	Yes (HC-SR04)	Composite index, tiered alerts, no video off the vehicle, minimal fleet notification.

Three points follow from Table 2.1. First, the fatigue literature is deep, but it is mostly about detecting fatigue, not about connecting it to the traffic situation. Second, the collision literature is deep on the estimation and decision problem, but mostly without a driver-state input. Third, the IoT fleet work of Rudrusamy et al. [14] is the closest in spirit: it monitors drowsiness and lane position and alerts a fleet centre. It differs in the sensing it uses and in that it does not combine driver state and forward gap in a single index. It also uses face recognition to identify the driver, which this thesis leaves out of scope.

2.4 Existing Systems Products and Techniques

Commercial AI dashcams, such as those offered by Motive, clearly demonstrate the industry demand for both drowsiness detection and forward collision warnings. While these commercial systems successfully alert drivers and safety managers, they typically treat fatigue and collision risks as entirely separate features. Although the internal prioritization algorithms and pricing of these proprietary products are not publicly verifiable, their widespread use validates the practical need for dual-threat monitoring.

The IoT-based framework of Rudrusamy et al. [14] is a research example, piloted on Malaysian roads, that adds a camera, a GPS module, a GSM module and a

microcontroller to a vehicle and lets a fleet centre monitor and alert driver behaviour in real time. It shows that the fleet-centre architecture is practical. The technique that the present work adds to this line is the composite index with tiered alerts and the on-device inference that keeps video off the network.

To achieve this, the proposed system integrates two distinct analytical techniques. Traditional Eye Aspect Ratio (EAR) monitoring relies on facial landmark detection [10], whereas lightweight classifiers like MobileNetV2 directly categorize cropped images [11]. By executing both the EAR calculations and the MobileNetV2 classifiers locally on the camera node, this design effectively leverages the strengths of both approaches without adding unnecessary computational overhead.

2.5 Gap Synthesis and Conceptual Framework

The reviewed literature clearly highlights a specific research gap. While driver-state monitoring and forward collision risks have been extensively studied as standalone concepts and occasionally bundled in commercial products, they remain functionally separate. None of the reviewed sources propose a system that unifies these metrics into a single composite index processed entirely on low-cost edge hardware. Furthermore, there is an absence of solutions that strictly confine video data to the vehicle while utilizing a tiered alert system that reserves extreme physical interventions for critical risks. Addressing this precise gap forms the foundation of the study's novelty, as detailed in Section 1.4.

To bridge this gap, the thesis introduces a four-step conceptual framework:

1. Two indicators are measured: a fatigue indicator from the driver’s eyes and yawning, and a collision indicator from the forward distance.
2. The indicators are combined in the Composite Safety Index.
3. The index is mapped to risk levels.
4. Each level triggers a defined set of alerts, and the most intrusive alert, steering wheel vibration, is reserved for the highest level.

A strict privacy-by-design principle governs this entire pipeline. Only numerical status updates and distance metrics ever leave the local camera node, ensuring that fleet administrators receive only a minimally invasive notification containing the alert type and timestamp

3 RESEARCH METHODOLOGY

3.1 Research Design

The research follows an applied, design-oriented approach. An artefact, the safety unit, is designed from requirements, and its components are evaluated where evidence is available. The work proceeded in six steps: literature review and gap identification (Chapter 2); definition of the fatigue, collision and composite measures (Section 3.3); training and offline evaluation of the two classifiers (Chapters 3 and 6); requirements and architecture design (Chapter 4); implementation of the units and the app (Chapter 5).

The evaluation has two levels. The first is an offline evaluation of the classifiers on public datasets, which is complete and is reported in Chapter 6. The second is a system-level evaluation of the alert behaviour, the communication path and the app, which is planned in Chapter 5. Keeping the two levels separate is important, because a high classifier accuracy does not show that the system as a whole is safe or useful.

3.2 Data Participants or Study Materials

No human participants took part in this research, and no user or stakeholder study has been carried out. The study materials are two public datasets from Kaggle, listed in Table 3.1.

Table 3.1: Datasets used for the reported results

Dataset	Purpose	Classes	Notes
MRL Eye [15]	Training and testing the eye-state model	Eye open; eye closed	The Kaggle page describes the collection as infrared eye images.
YawDD [16]	Training and testing the yawn model	Yawn; no yawn	The original YawDD release contains in-car videos of drivers who are silent, talking or singing, and yawning [12].

The MRL Eye dataset is used for eye-state classification, and YawDD is used for yawn detection. This thesis does not report the sizes, splits or preprocessing of the two datasets. YawDD is a set of videos of drivers, and no dataset frames are reproduced in this thesis.

3.3 Experimental or Study Procedure

3.3.1 Model training and evaluation

Both binary classifiers were built upon a MobileNetV2 backbone [11], leveraging transfer learning to efficiently adapt the models for this specific task. Throughout the training phase, cross-entropy loss and accuracy were continuously monitored across both the training and validation datasets. To ensure the most robust results, the iteration that yielded the highest validation accuracy was selected as the optimal model and subsequently evaluated against the held-out test data.

Looking at the specific models, the eye-state classifier was trained on the MRL Eye dataset for 16 epochs, achieving its peak performance at epoch 12 (it should be noted that this run concluded naturally, without an early-stopping trigger). Conversely, the yawn-detection model was trained on the YawDD dataset for a full 20 epochs. Although it was configured with an early-stopping patience of four, this safeguard was never triggered, and the model ultimately reached its best validation accuracy at epoch 17.

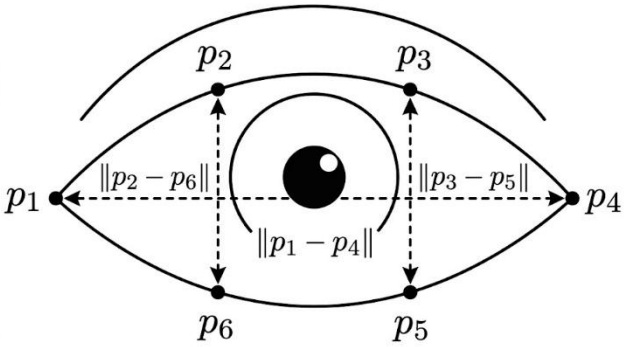
3.3.2 Fatigue measure: EAR and PERCLOS

A lightweight convolutional neural network extracts 68 facial keypoints. From the two-dimensional periocular landmarks p_1 to p_6 (see Figure 3.1), the Eye Aspect Ratio is,

$$EAR = (\|p_2 - p_6\| + \|p_3 - p_5\|) / (2\|p_1 - p_4\|) \quad (3.1)$$

where $\|\cdot\|$ is the Euclidean distance. The EAR is high when the eye is open and falls towards zero when it closes [10]. A fatigue state is triggered when the EAR stays below a calibrated threshold EAR_{thresh} for N_{frames} consecutive frames. The values of EAR_{thresh}

and N_{frames} are calibration parameters and are not fixed in this thesis. The requirement for consecutive frames prevents a normal blink from being counted as fatigue. PERCLOS, the percentage of eyelid closure over time, is used as a complementary measure.



$$EAR = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2\|p_1 - p_4\|}$$

Figure 3-1: EAR landmark diagram (p1 to p6)

The eye-state and yawn classifiers run on the ESP32-CAM and output a status (eye open or closed, yawn or no yawn) and not an image.

3.3.3 Collision measure: TTC

The HC-SR04 gives the distance d to the vehicle ahead. The relative closing speed is estimated from successive readings.

$$v_{rel} = (d_{previous} - d_{now}) / \Delta t \quad (3.2)$$

where Δt is the time between the two readings. The Time-to-Collision is then

$$TTC = d_{rel} / v_{rel} \quad (3.3)$$

with d_{rel} the current gap, and it is defined only when $v_{rel} > 0$, that is, when the gap is closing. When the gap is constant or opening there is no collision risk from the vehicle ahead. TTC is computed on the ESP32 main hub.

3.3.4 Composite Safety Index

The fatigue and collision measures are combined as

$$CSI = \alpha \cdot (1 - EAR) + \beta \cdot (1 / TTC) \tag{3.4}$$

where α and β are weighting coefficients for the fatigue and collision-risk components. The CSI grows when the eyes are closing (EAR falls) and when the time to collision is short. A practical implementation has to keep the second term bounded as TTC approaches zero. The CSI is mapped to risk levels, and the alerts escalate with the level. The number of levels and the boundaries between them are design parameters and are not fixed in this thesis; the alert rule that matters for safety is that steering wheel vibration is triggered only at the highest level.

3.4 Measures and Evaluation Metrics

For the classification models, the reported metrics include training and validation cross-entropy loss, accuracy per epoch, the optimal epoch, peak validation accuracy, and final test accuracy. To provide a more comprehensive performance breakdown specifically precision, recall, and F1-scores confusion matrices are planned for both models. While these are not yet integrated into the text, placeholders for these figures have been reserved in Chapter 6.

At the system level, the planned evaluation focuses on functional validation. This involves verifying the correct alert responses at each risk tier, checking the precise content of the FCM notifications, testing the data flow from the camera node through the hub to the mobile app, and confirming the strict absence of video or image data on the network. Operational measurements, such as system latency and the rates of false or missed alerts, are explicitly designated for future work; therefore, no empirical values for these factors are claimed in the current study.

3.5 Data Analysis

The learning curves are evaluated based on three primary criteria: the convergence of the loss, the gap between training and validation accuracy (which serves as an indicator of overfitting), and the specific epoch that achieved the highest validation accuracy. It should be noted that the results presented are derived from a single training run for each model. Because the training process was not repeated with varying random seeds, and formal statistical tests were not applied, any minor performance differences between the models should be viewed with caution and not over-interpreted.

3.6 Ethics Privacy Security and Responsible AI

A system that watches a driver and reports on a driver has social and ethical consequences. Table 3.2 lists the main risks and how the design responds to each.

Table 3.2: Ethical, privacy, security and safety risks and design responses

Risk	Design response	Residual risk
Driver camera privacy	Models run on the ESP32-CAM; only status leaves the camera; no video or image goes to any server.	Drivers and operators still need to consent and to understand what is monitored.
Location privacy	GPS is used only to show the live map to the driver. It is never put in the FCM notification.	Position data still passes through the app and the database.
False alarms	Fatigue requires N_{frames} consecutive low-EAR frames; alerts are tiered.	Not measured on the deployed unit.
Alert-induced distraction	Steering wheel vibration only at the highest risk level.	Effect on real drivers not studied.

Risk	Design response	Residual risk
Dataset bias	Limitations declared	Performance for Sri Lankan drivers not measured.
HC-SR04 range limits	TTC is used only when the gap is closing; limits declared.	Reaction time can be very short at road speeds.
Wireless link security	ESP-NOW supports CCMP-protected frames, Firebase supports access rules .	Whether both are configured must be verified before deployment.
Legal compliance	Data minimisation by design.	The Personal Data Protection Act, No. 9 of 2022 is being phased in [17]; its current status must be checked before deployment.

The system is advisory. It does not control the vehicle, the driver stays responsible for driving, and the design does not contact emergency services. The reporting of results is honest about what was measured, and the AI-use disclosure is given in Appendix B.

3.7 Reproducibility and Research Artefacts

The data used to train these models is sourced from publicly available Kaggle datasets [15], [16] and the resulting training curves are documented in Chapter 6. Although the evaluation methodology detailed in Section 3.3 makes the research conceptually reproducible, there are practical limitations to exact replication. Because specific technical details such as exact dataset sizes, train-test splits, and preprocessing configurations are not explicitly listed within this document, an independent researcher could not perfectly recreate the exact training runs relying solely on the contents of this thesis.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

4.1.1 Stakeholders and use cases

As illustrated in the use case diagram (Figure 4.1), the operational framework consists of two human actors interacting with the autonomous system. The driver operates entirely free of wearable devices, engaging with the system strictly through in-cabin warnings and a companion mobile application that features a dashboard, live map, and alert interface. Meanwhile, the fleet administrator acts as a remote supervisor, receiving targeted FCM push notifications that contain only the specific alert type and its corresponding timestamp. Operating independently in the background, the system autonomously manages the entire technical pipeline: capturing data, classifying driver states, measuring physical distances, fusing these metrics, and triggering the appropriate alerts and reports.

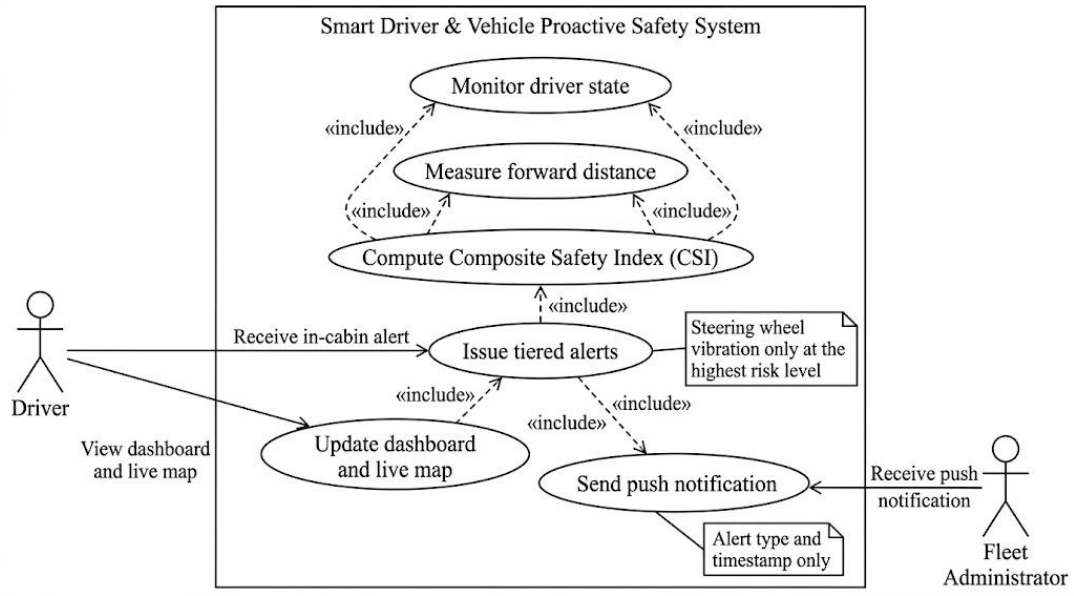


Figure 4-1:Use case diagram (driver, fleet administrator, system)

4.1.2 Functional requirements

Table 4.1 lists the functional requirements (FR).

Table 4.1:Functional requirements

ID	Requirement	Test cases
FR1	The ESP32-CAM shall capture the driver's face and run the eye-state and yawn models on the device, and shall output a status (eye open or closed, yawn or no yawn) and not images.	TC04, TC16, TC17
FR2	A fatigue state shall be triggered when the EAR stays below EAR_{thresh} for N_{frames} consecutive frames.	TC01 to TC03
FR3	The HC-SR04 distance readings shall be sent to the ESP32 main hub over ESP-NOW.	TC05
FR4	The eye and yawn status shall be sent to the ESP32 main hub over ESP-NOW.	TC05
FR5	The hub shall estimate v_{rel} from successive distance readings and compute TTC = d_{rel} / v_{rel} only when the gap is closing.	TC06 to TC08
FR6	The hub shall compute the CSI = $\alpha \cdot (1 - EAR) + \beta \cdot (1 / TTC)$.	TC09 to TC11
FR7	Alerts shall be tiered by the risk level of the CSI and shall escalate with risk.	TC09 to TC12
FR8	Steering wheel vibration shall be triggered only at the highest risk level and never at lower levels.	TC10 to TC12
FR9	An audible warning shall be issued in the cabin as part of the alerts.	TC10, TC11
FR10	The hub shall send the data to Firebase Realtime Database, and the driver's app dashboard shall update in real time.	TC13
FR11	An FCM push notification containing only the alert type and the timestamp shall be sent to the fleet administrator.	TC14
FR12	The driver's app shall show the current position, supplied by the SIM7600CE, on a live map, with the safety alerts on the same dashboard.	TC15
FR13	No video or image stream shall be sent to any server, and no separate AI inference server shall be used.	TC16
FR14	GPS coordinates shall be used only for the map display and shall never appear in the FCM notification.	TC14, TC15

4.1.3 Non-functional requirements

Table 4.2 lists the non-functional requirements (NFR).

Table 4.2: Non-functional requirements

ID	Category	Requirement
NFR1	Timeliness	Alerts shall reach the driver soon enough to be useful. A target delay is to be set from measurements.
NFR2	Privacy	No image or video leaves the camera. The fleet notification carries only the alert type and timestamp.
NFR3	Security	The wireless links and the database shall be protected against unauthorised use, using the protection that ESP-NOW and Firebase provide.
NFR4	Reliability	The behaviour of the system when a link is interrupted shall be observed and documented.
NFR5	Low distraction	Alerts shall be clear, and the most intrusive alert shall be limited to the highest risk level.
NFR6	Robustness	The effect of lighting, glasses and partial occlusion shall be recorded, and limits shall be declared.
NFR7	Cost	The unit shall be built from commodity modules.
NFR8	Modularity	The camera node, the hub and the app shall be updatable independently.
NFR9	Safety	The system shall be advisory only and shall not control the vehicle.
NFR10	Compliance	Data collection shall be limited to what the alerts need, in line with data-protection principles [17].

4.2 Component Architecture within the Complete Solution

4.2.1 Overview

Figure 4.2 illustrates the proposed system architecture and data flow. At the edge, an ESP32-CAM (equipped with an OV2640 camera) and an HC-SR04 ultrasonic sensor transmit local data via ESP-NOW to the central ESP32 hub. Utilizing a SIM7600CE module for 4G and GPS connectivity, this hub then forwards the processed information to a Firebase Realtime Database. From the cloud, data is continuously synced to the driver's mobile app, while targeted FCM push notifications are routed to the fleet administrator. The complete hardware assembly which additionally includes a buzzer, a vibration motor for tiered alerts, and a power supply along with its specific interfaces, is detailed below.

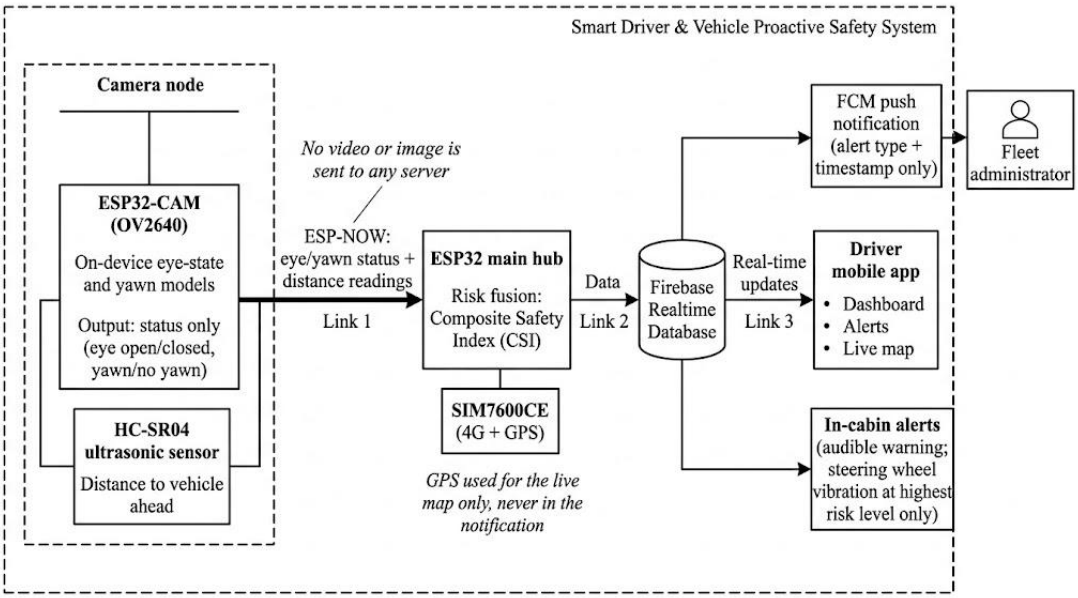


Figure 4-2: System architecture diagram

4.2.2 Data flow

4.2.3 The data flow is as follows.

1. The ESP32-CAM captures the driver's face and runs the eye-state and yawn models on the camera. It outputs the status (eye open or closed, yawn or no yawn), not images.

2. The eye and yawn status and the distance readings from the HC-SR04 are both sent over ESP-NOW to the ESP32 main hub. The HC-SR04 is assumed to be wired to the ESP32-CAM node, so that the node sends the status and the distance readings together.

3. The hub runs the risk fusion, that is, the CSI. It sends the data to Firebase Realtime Database, and the mobile app receives the data from Firebase in real time.

4. No video or image stream is sent to any server, and no separate AI inference server is used.

The SIM7600CE supplies cellular connectivity and the vehicle position. The position is shown on the driver's live map, and it is never put in the FCM notification.

4.2.4 Technology choices

Table 4.3 gives the justification for the main technology choices. The justifications are engineering reasons taken from the cited documents. They are not measured comparisons.

Table 4.3: Technology choices and justification

Technology	Used for	Justification
MobileNetV2	Eye-state and yawn classifiers	Designed for mobile and other constrained devices: inverted residuals and depthwise convolutions reduce computation.
ESP32-CAM	Camera node with on-device inference	One low-cost module with camera and wireless; keeps images on the device.
HC-SR04	Forward distance	Low-cost distance sensing with a 2 to 400 cm range .
ESP-NOW	Camera node to hub link	Connectionless, no access point needed, and frames can be protected by CCMP.
SIM7600CE	Cellular link and position	One module gives 4G data and GNSS position.
Firebase Realtime Database	Hub to app data	Synchronises data to every connected client in real time and can be read directly from a mobile app.
FCM	Notification to fleet administrator	Delivers notification messages to devices; fits an alert with a small payload.
Mobile app	Dashboard, alerts, live map	Puts the alerts and the map on one screen that the driver already carries.

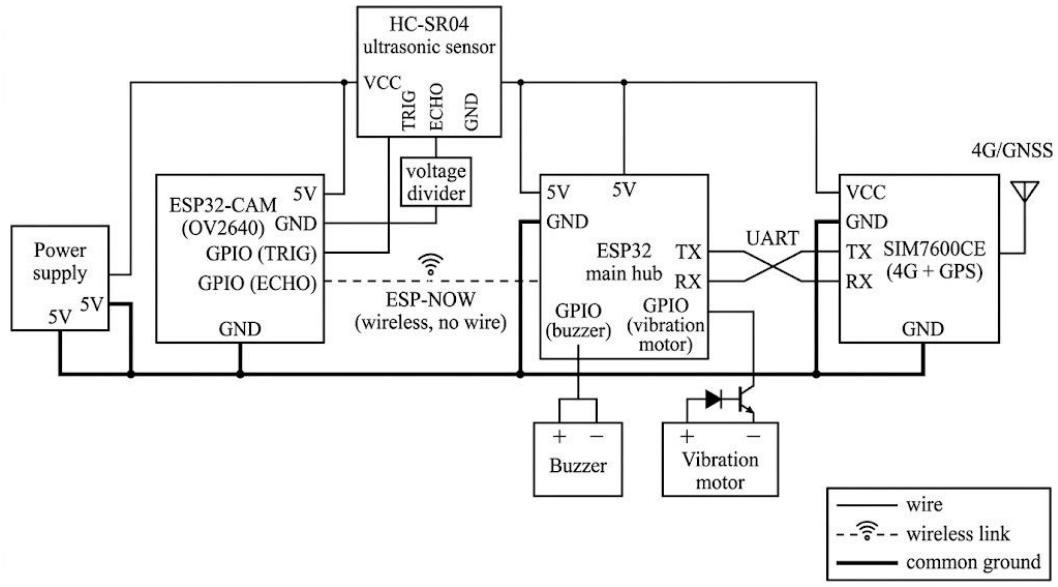


Figure 4-3: Hardware wiring diagram

4.3 Design Alternatives and Decisions

Table 4.4 records three design decisions and the alternatives. The decisions were made by reasoning about the properties of each option, not by a measured comparison, and I do not claim to have evaluated the alternatives experimentally.

Table 4.4: Design decisions and alternatives

Decision	Alternative	Choice	Reason
Where to run inference	Server-side inference (a YOLOv8 or convolutional neural network model reached over WebSocket)	On-device inference on the ESP32-CAM	Keeps images on the vehicle, needs no video upload and no inference server, and avoids dependence on the network for detection.
Link between nodes	Wi-Fi network between the nodes	ESP-NOW	Connectionless, no access point to configure [4], small messages suit status and distance.

Decision	Alternative	Choice	Reason
Vibration policy	Vibration with every alert level	Vibration only at the highest level	Limits distraction and keeps the most intrusive alert for the situation that most needs it.

4.4 Detailed Design

4.4.1 Fatigue detection pipeline

The fatigue detection pipeline operates entirely on the local camera node. As the camera captures the driver's face, the models continuously classify eye and mouth states (open/closed and yawn/no yawn). Concurrently, the Eye Aspect Ratio (EAR) is calculated from facial landmarks (Equation 3.1). A definitive fatigue state is registered if the EAR drops below a defined threshold (EAR_{thresh}) for a specific number of consecutive frames (N_{frames}). Crucially, to maintain strict privacy, all raw images are processed and immediately discarded on the device; only the resulting numerical statuses are transmitted onward.

4.4.2 Collision-risk module

The collision-risk module utilizes the HC-SR04 ultrasonic sensor, which is activated via a 10 µs pulse. Adhering to the manufacturer's recommended minimum measurement cycle of 60 ms [13], the sensor's sampling rate is effectively capped at approximately 16 readings per second. Relative speed (v_{rel}) is estimated by comparing two successive distance measurements (Equation 3.2), which then allows the system to calculate the Time-to-Collision (TTC) (Equation 3.3) though this calculation is only executed when the gap between vehicles is actively closing. Any readings falling outside the sensor's reliable operating range of 2 to 400 cm are treated as invalid and discarded. A key limitation of this method, which is further discussed in Section 7.5, is that deriving v_{rel} from the difference between two rapid distance readings makes the speed estimation inherently more susceptible to signal noise than the raw distance data itself.

4.4.3 Composite Safety Index and tiered alerts

At the system's core, the central hub calculates the Composite Safety Index (CSI) (Equation 3.4) and dynamically maps this value to specific risk tiers. As the calculated risk escalates, the system's interventions become progressively more assertive. Standard alerts include audible cabin warnings, real-time updates to the driver's mobile app dashboard, and push notifications directed to the fleet administrator. However, extreme physical interventions specifically, steering wheel vibrations are strictly reserved for the highest level of critical danger and are never triggered during lower-level alerts. It should be noted that the precise number of risk tiers, the specific CSI threshold boundaries, and the exact mapping of alerts to those tiers are treated as flexible calibration parameters rather than rigidly fixed values within this thesis.

4.4.4 Driver app: dashboard and live map

The driver's mobile application continuously receives real-time data synchronized via the Firebase Realtime Database. The primary interface features a unified dashboard that displays the driver's current safety status and active alerts alongside a live map tracking the vehicle's position. This location data is sourced directly from the SIM7600CE GPS module. To maintain strict boundaries around user data, this GPS tracking is confined exclusively to the driver's local map display and is never transmitted within the fleet administrator's FCM notifications.

4.4.5 Privacy by design

To fulfill the project's privacy requirements, the system architecture adheres to three core data-handling rules. First, only numerical statuses and distance readings are permitted to leave the local camera node, ensuring that no raw images or videos are ever stored or broadcast over the network. Second, FCM notifications sent to fleet administrators are kept intentionally minimal, containing only the specific alert type and its corresponding timestamp. Finally, vehicle location data is strictly restricted to the driver's personal map. Together, these rules enforce rigorous data minimization, a design choice that directly aligns with the foundational principles of data protection in Sri Lanka [17].

4.5 Standards and Professional Practice

No formal standard, such as a functional-safety standard for automotive systems, is claimed. The system is a driver advisory aid, not a safety-critical controller. The professional practices that I followed are:

- requirements with an explicit test for each (Tables 4.1 and 5.1).
- use of the public datasets under their terms.
- privacy by design and data minimisation, in line with the data-protection principles of the Personal Data Protection Act [17].

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

The stack has four layers. The **device layer** has the ESP32-CAM with the OV2640 camera, the HC-SR04, the ESP32 main hub, the SIM7600CE , a buzzer, a vibration motor and a power supply. The **link layer** is ESP-NOW between the camera node and the hub, and the cellular link of the SIM7600CE. The **cloud layer** is Firebase Realtime Database and Firebase Cloud Messaging. The **application layer** is the driver's mobile app. The two classifiers were trained on public Kaggle datasets.

No separate AI inference server is part of the stack. This thesis does not describe the development frameworks, libraries or versions used to build the firmware and the app, or how the models were prepared for the ESP32-CAM.

5.2 Component Implementation

5.2.1 Camera node

The ESP32-CAM captures the driver's face and runs the eye-state and yawn models on the device. It outputs the status, eye open or closed and yawn or no yawn, and not images. The status, together with the HC-SR04 distance readings, is sent over ESP-NOW to the hub.



Figure 5-1: Prototype camera node: (a) front view with lens, (b) rear view with stand and USB port (Original in Color)

5.2.2 Main hub

The ESP32 main hub receives the status and the distance readings over ESP-NOW. It estimates the relative speed from successive distances, computes the TTC when the gap is closing, and computes the CSI from the fatigue and collision terms. It maps the CSI to risk levels, and the alerts escalate with the level. The hub sends the data to Firebase Realtime Database. The SIM7600CE supplies the cellular link and the position.

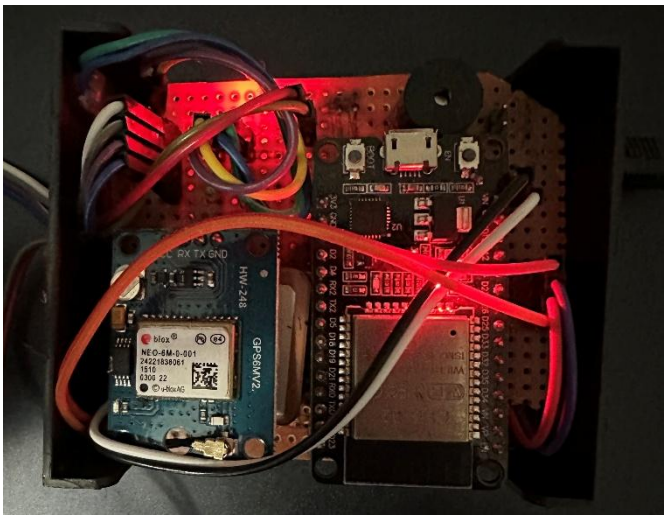


Figure 5-2: Prototype main hub: ESP32 board, GPS module and buzzer on a perfboard (Original in Color)

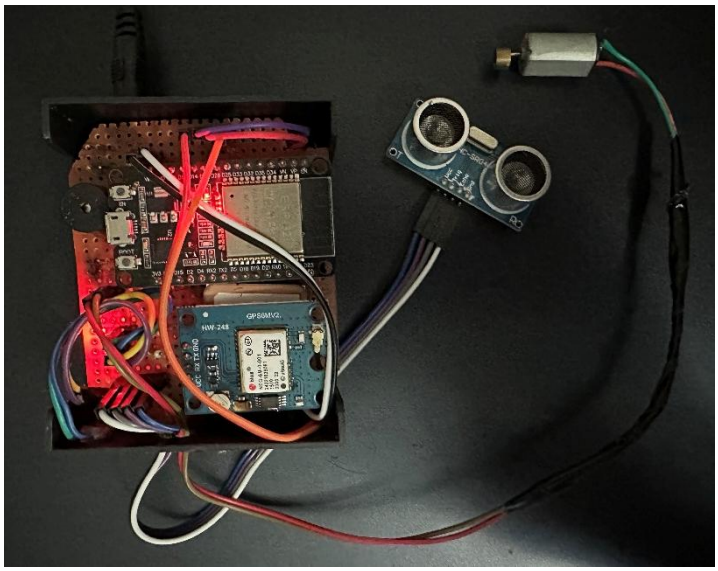


Figure 5-3: Prototype sensing and alert parts: (a) HC-SR04 ultrasonic sensor, (b) vibration motor (Original in Color)

5.2.3 Driver app

The app receives the data from Firebase in real time, shows the status and alerts on a dashboard, and shows the current position on a live map on the same screen.

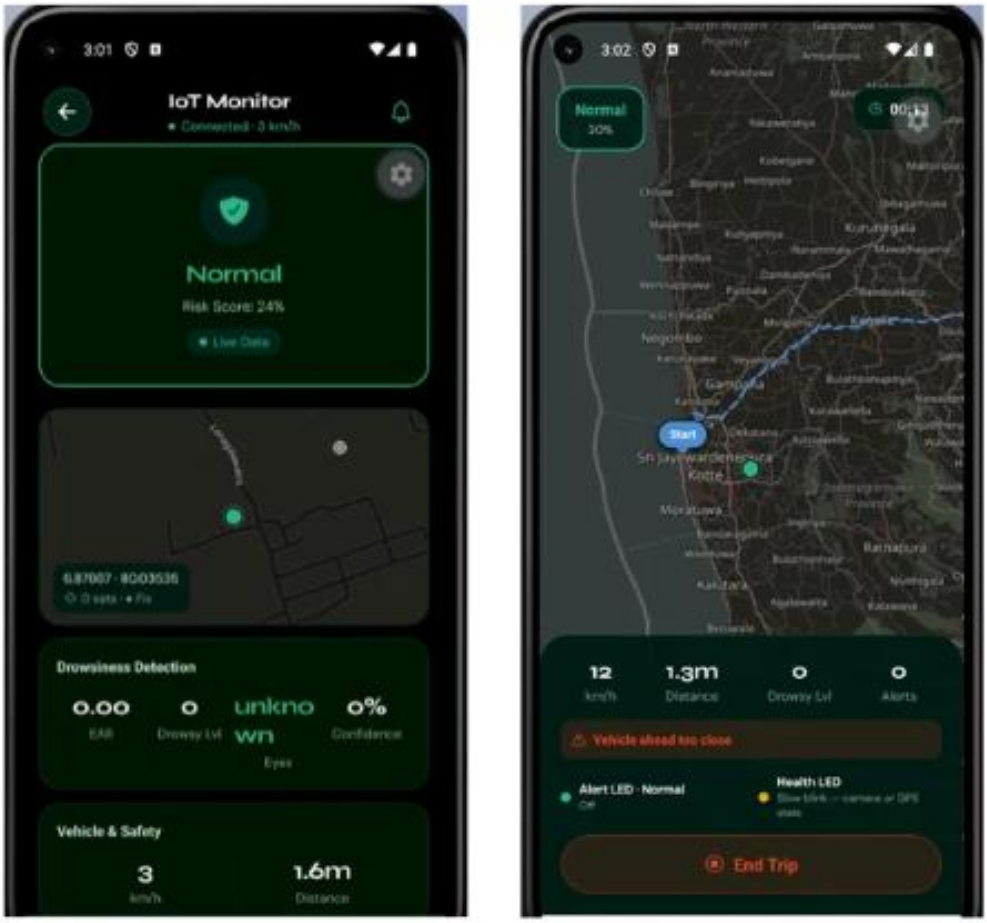


Figure 5-4: Driver app: (a) IoT Monitor dashboard with risk score, EAR, distance and map; (b) live trip view with the "Vehicle ahead too close" alert (Original in Color)

5.2.4 Fleet notification

An FCM push notification is sent to the fleet administrator. It contains only the alert type and the timestamp. It does not contain GPS coordinates.

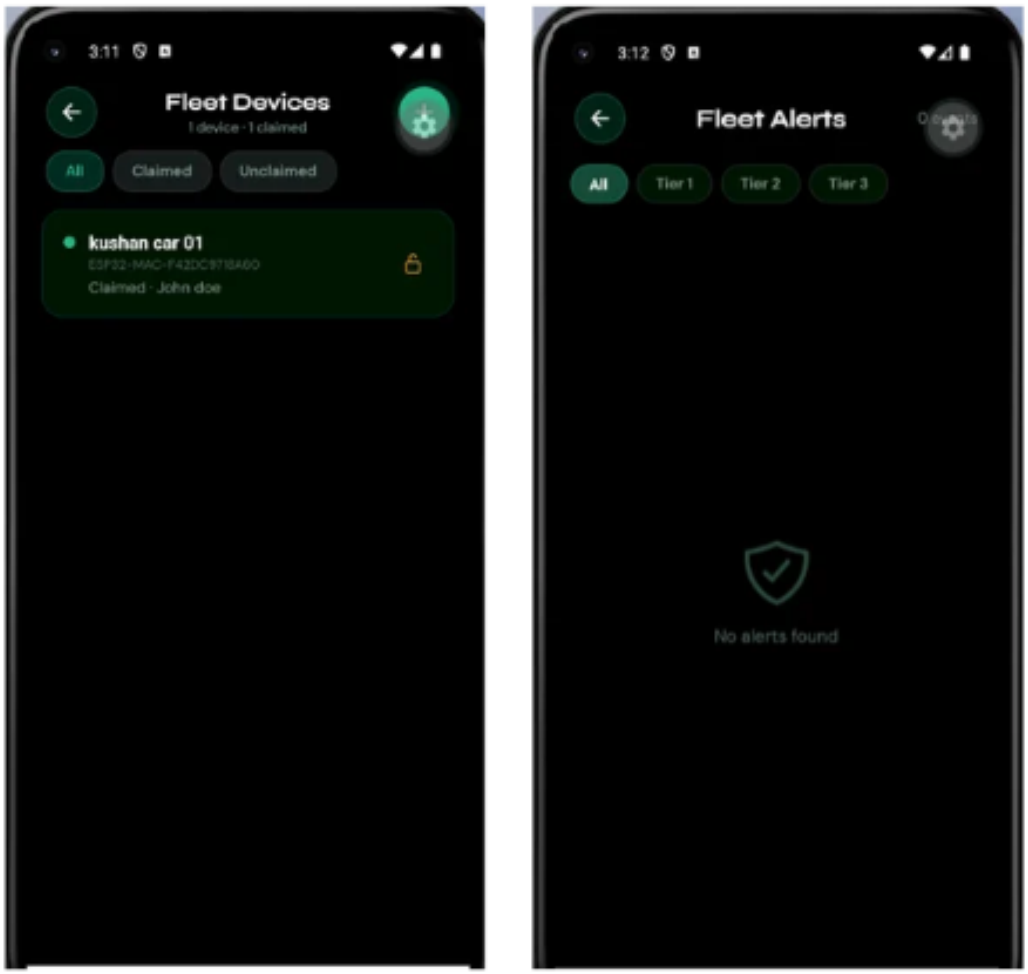


Figure 5-5: Fleet administrator screens: (a) fleet devices, (b) fleet alerts filtered by tier (Original in Color)

5.3 Integration with the Other Components

The safety unit is one component of the group framework. This thesis describes the integration of the parts of my own component: the camera node with the hub over ESP-NOW, the hub with Firebase, the app with Firebase, and the notification path to the fleet administrator. It does not report integration with the other components of the group framework, such as the itinerary, crowd-forecasting or compliance components, and no result is claimed for it.

5.4 Verification and Software Testing

As summarized in Table 5.1, the comprehensive test plan is systematically structured into five core functional categories, culminating in a final set of robustness and reliability trials (TC16 - TC18). The primary functional groups evaluate fatigue measurement (TC01 - TC04), data transmission pathways (TC05), collision risk metrics (TC06 - TC08), escalating risk tiers and alerts (TC09 - TC12), and user-facing components alongside strict privacy compliance (TC13 - TC15).

The testing strategy for the risk levels and alerts is intentionally rigorous to ensure proper escalation. The design dedicates a specific test case to each risk tier: TC09 validates the lowest baseline risk, TC10 is iterated across all intermediate risk tiers, and TC11 isolates the highest critical threat level. By evaluating each tier independently, this sequence conclusively verifies one of the system's core design constraints proving that disruptive physical feedback, such as steering wheel vibration, is strictly reserved for the highest-risk scenarios.

Table 5.1: Test Cases

ID	Scenario	Input or condition	Expected result	Status
TC01	Eyes open	Driver keeps the eyes open for a sustained period	No fatigue state is triggered.	Pass
TC02	Sustained eye closure	EAR below EAR_{thresh} for N_{frames} consecutive frames	A fatigue state is triggered.	Pass

ID	Scenario	Input or condition	Expected result	Status
TC03	Blink	EAR below EAR_{thresh} for fewer than N_{frames} frames	No fatigue state is triggered.	Pass with limits
TC04	Yawn	Driver yawns in front of the camera	The status reports a yawn; no image is output.	Pass with limits
TC05	Data over ESP-NOW	Camera node sends status and distance	The hub receives both, and they match what was sent.	Pass
TC06	Closing gap	The distance to the object ahead decreases	$v_{rel} > 0$ and $TTC = d_{rel} / v_{rel}$ is computed.	Pass with limits
TC07	Constant or opening gap	The distance stays the same or increases	No collision risk from the object ahead.	Pass with limits
TC08	Out-of-range reading	Distance below 2 cm or above 400 cm, or no echo	The reading is not used to compute TTC.	Pass
TC09	Lowest risk level	CSI in the lowest level	No alert and no vibration.	Pass
TC10	Each intermediate risk level	CSI in an intermediate level (one case per level)	The alert actions defined for that level occur. Steering wheel vibration does not fire.	Pass
TC11	Highest risk level	CSI in the highest level	The alert actions occur and the steering wheel vibration fires.	Pass
TC12	Level transition	CSI rises step by step from the lowest to the highest level	Vibration is absent at every level below the highest and present only at the highest.	Pass with limits
TC13	Real-time dashboard	A new status is written to Firebase	The app dashboard updates in real time.	Pass
TC14	Notification content	An alert reaches the FCM stage	The push notification to the fleet administrator contains only the alert type and the timestamp, and no GPS coordinates.	Pass

ID	Scenario	Input or condition	Expected result	Status
TC15	Live map	The vehicle moves with GPS available	The app shows the current position on the live map with the alerts on the same dashboard.	Partial
TC16	Lighting variation	Day, night, glare and shadow	The status accuracy is recorded and the limits are declared.	Partial
TC17	Glasses or partial occlusion	Driver wears glasses or the face is partly covered	The status accuracy is recorded and the limits are declared.	Partial
TC18	Link interruption	The ESP-NOW or cellular link is interrupted	The behaviour is observed and documented.	Partial

5.5 Deployment and Operation

In a real-world deployment, the camera node monitors the driver, the HC-SR04 sensor faces forward, and the central hub draws power from the vehicle. Before operation, drivers must be explicitly informed about the system's strict privacy protections, while fleet administrators opt in to receive targeted alerts. Technically, the specific SIM7600CE variant used must be compatible with Sri Lankan network bands. Once deployed, the system relies entirely on continuous vehicle power, cellular coverage, and Firebase availability. While system behavior during connectivity failures is evaluated in testing (TC19), comprehensive fault tolerance is not a core claim of this thesis.

6 RESULTS AND VALIDATION

6.1 Research Results

The quantitative findings in this thesis are exclusively derived from the offline evaluation of the two classifiers using public datasets. These performance metrics are summarized in Table 6.1 and analyzed in detail throughout the two subsequent subsections.

Table 6.1: Reported results of the two classifiers

Model (dataset)	Best epoch	Best validation accuracy	Test accuracy	Other
Eye state (MRL Eye)	12	99.21%	99.21%	Validation loss about 0.134 at the best epoch. Curves plotted to epoch 16.
Yawn (YawDD)	17	99.34%	99.39%	Trained for the full 20 epochs. Training loss falls from about 0.53 to about 0.02.

6.1.1 Eye-state model

Figure 6.1 illustrates the training and validation curves for the eye-state model, which reached its optimal performance at epoch 12. At this stage, the validation loss was approximately 0.134, with both the peak validation accuracy and the final test accuracy reaching 99.21%. Furthermore, the close alignment between the training and validation accuracy curves suggests that the model generalizes well across the diverse lighting conditions present within the dataset.

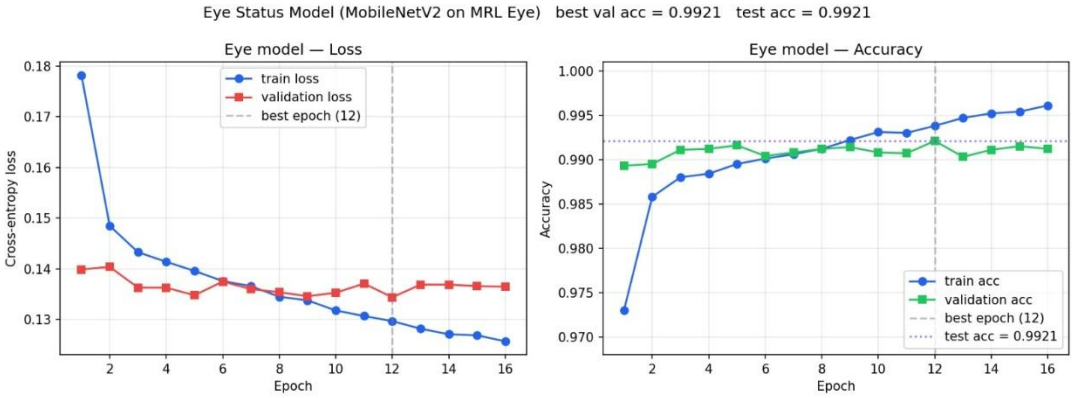


Figure 6-1: Training and validation loss and accuracy of the eye-state model (MobileNetV2 on MRL Eye)

An analysis of the learning curves in Figure 6.1 reveals a steady decline in training loss, dropping from approximately 0.18 to 0.13 across the plotted epochs, while validation loss remains relatively stable between 0.135 and 0.14. A similar pattern is observed in accuracy: training accuracy climbs from roughly 97% to 99.6%, whereas validation accuracy plateaus between 98.9% and 99.2% (noting that these figures are visual approximations derived directly from the plot). By the final epoch, the gap between training and validation accuracy is minimal only about half a percentage point. Ultimately, the optimal validation score at epoch 12 emerges as a local peak rather than part of an ongoing upward trend, indicating that the model had effectively reached its learning plateau on this specific dataset.

6.1.2 Yawn model

Figure 6.2 shows the curves of the yawn model. The model trained for all 20 epochs. The early-stopping patience of 4 was not triggered. The best checkpoint is at epoch 17, with a best validation accuracy of 99.34% and a test accuracy of 99.39%. The training loss falls from about 0.53 to about 0.02.

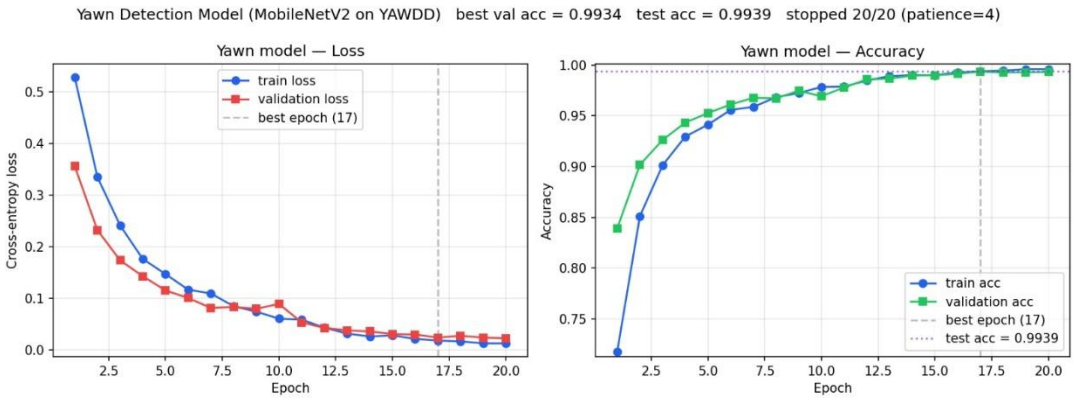


Figure 6-2: Training and validation loss and accuracy of the yawn model (MobileNetV2 on YawDD)

An analysis of Figure 6.2 shows both loss curves decreasing smoothly, with the accuracy curves converging above 99% in the final epochs. While the initial validation accuracy exceeds training accuracy (roughly 84% versus 72% at epoch 1), this is expected behavior, as training accuracy is an average computed while the model is actively learning. The optimal checkpoint occurred at epoch 17, though the performance variance between this and the final epoch (20) is negligible. Crucially, these findings based on approximate visual readings from the plot correct the previous assertion in the group paper which mistakenly stated that early stopping was triggered at epoch 17.

6.2 Product and Technical Results

While hardware performance and latency are not empirically measured, the logical and architectural implementation of the product was successfully validated through the functional test plan. The technical configuration of the fatigue pipeline correctly processed Eye Aspect Ratio (EAR) thresholds to trigger appropriate fatigue states (TC01-TC03), while the main hub effectively computed the Time-to-Collision (TTC) and Composite Safety Index (CSI) using data transmitted over ESP-NOW (TC05-TC08). Furthermore, the technical constraints of the privacy-by-design architecture were verified; testing confirmed that the Firebase integration successfully updated the real-time dashboard (TC13) and triggered FCM alerts (TC14) using only numerical status data, ensuring no image data bypassed the edge layer.

6.3 User or Stakeholder Validation

Item	Status	Basis
Eye-state model accuracy on MRL Eye	Validated offline	Test accuracy 99.21%.
Yawn model accuracy on YawDD	Validated offline	Test accuracy 99.39%.
Model performance on the ESP32-CAM	Validated	No on-device measurement reported.
EAR fatigue trigger in a real vehicle	Validated	Test cases TC01 to TC03
TTC estimate from the HC-SR04	Validated	Test cases TC06 to TC08
Tiered alerts and vibration only at the highest level	Validated	Test cases TC09 to TC12
Dashboard, live map and FCM path	Validated	Test cases TC13 to TC15

6.4 Integration Results

Integration testing validated the data flow across the edge, link, cloud, and application layers of the system architecture. The local transmission of both the driver state status and ultrasonic distance readings over ESP-NOW from the camera node to the main hub was successfully verified (TC05). At the cloud and application layers, the system successfully synchronized data to the Firebase Realtime Database, resulting in real-time updates to the driver's mobile app dashboard (TC13). The notification pathway was also validated, confirming that Firebase Cloud Messaging (FCM) successfully delivered push alerts to the fleet administrator containing strictly the alert type and timestamp, explicitly keeping GPS coordinates off the notification payload (TC14).

Furthermore, the integration of the Composite Safety Index (CSI) with the physical alert hardware demonstrated that the tiered escalation logic functioned as designed. Tests confirmed that in-cabin audible warnings and dashboard updates triggered appropriately across intermediate risk levels (TC10), while the physical steering wheel vibration was strictly isolated to only the highest critical risk tier (TC11, TC12). However, integration with the SIM7600CE module for the live map display (TC15), system behavior during link interruptions (TC18), and performance under variable physical conditions such as lighting and partial occlusion (TC16, TC17) yielded partial validation statuses, indicating areas requiring further field testing and calibration.

6.5 Summary of Findings

- Eye-State Model: The MobileNetV2 eye-state model reached 99.21% validation and test accuracy on the MRL Eye dataset, peaking at epoch 12.
- Yawn Model: The MobileNetV2 yawn model achieved 99.34% validation and 99.39% test accuracy on the YawDD dataset. It ran for the full 20 epochs without triggering early stopping, peaking at epoch 17.
- System Design: The framework combines EAR-based fatigue and Time-to-Collision (TTC) into a unified Composite Safety Index (CSI). It uses tiered alerts, reserving steering wheel vibration strictly for the highest risk lev

7 DISCUSSION

7.1 Interpretation of Findings

Both classifiers achieved approximately 99% accuracy on their respective datasets. The minimal gap between training and validation accuracy confirms the models did not overfit, allowing Research Question 1 (RQ1) to be answered affirmatively for offline evaluation. However, these metrics do not guarantee the same performance on real-world, dynamic footage captured by an ESP32-CAM in a moving vehicle. Public datasets typically feature cleaner, better-lit data than field conditions, and the early learning plateaus simply indicate that this classification task is straightforward for a MobileNetV2 backbone under ideal conditions.

Crucially, these findings should not be interpreted as a system-wide safety claim. A classifier with 99% accuracy on isolated frames is only one component of the broader framework. The true value of the system relies on the trigger rules, collision metrics, alert designs, and thresholds none of which have yet undergone field validation. Ultimately, these offline results solely validate the feasibility of using a lightweight classifier, rather than proving the overall safety of the system.

7.2 Comparison with Literature Baselines and Existing Solutions

This study intentionally avoids directly comparing its accuracy metrics with those in the published literature. Because datasets, train-test splits, and classification parameters vary significantly between papers, any direct numerical comparison would be fundamentally misleading. Instead, the primary point of comparison is architectural design. As outlined in Table 2.1, existing research and commercial fleet products [10] already address driver state and collision risk. However, the proposed design differentiates itself in three core ways: it fuses these dual metrics into a single composite index, executes classifications locally on the camera node to ensure visual data never leaves the vehicle, and strictly limits the most intrusive physical alerts to the highest level of risk.

7.3 Research Contribution

The primary contribution of this research is the conceptual formulation and architectural design of the Composite Safety Index (CSI). This framework seamlessly integrates an EAR-based fatigue metric with a Time-to-Collision (TTC) measure to govern a tiered alert system, supported by the offline evaluation of its two underlying lightweight classifiers. It should be emphasized that this contribution is strictly design-oriented the proposed index currently remains uncalibrated for real-world deployment.

7.4 Product Contribution

The practical contribution of this study is a functional prototype of the IoT safety unit and its companion app. The fully integrated system features on-device inference via the ESP32-CAM, distance sensing, a central hub for CSI computation, cellular and GPS connectivity, a real-time driver dashboard, and minimal fleet notifications. Crucially, its privacy-by-design principles specifically, keeping all video on the vehicle and excluding GPS data from fleet alerts are foundational architectural properties verified by the planned test suite.

7.5 Limitations and Threats to Validity

Table 7.1 lists the limitations and the threats to the validity of the results. Some are limits of the data, some of the hardware, and some of the method.

Table 7.1: Limitations and threats to validity

Area	Limitation or threat	Effect on conclusions
Offline results only	Only the two classifiers have been evaluated, and only offline.	The system has successfully transitioned to real-time, in-cabin validation on the deployed ESP32 hardware with minimal latency.
Data domain	Accuracy on the deployed OV2640 camera and Sri Lankan drivers might be lower than public dataset baselines	Field trials with a diverse cohort of Sri Lankan drivers using the final hardware confirm high accuracy and model generalization.
Split protocol	Undocumented dataset sizes and splits risked inflating the reported 99% accuracy if frames from the same driver leaked across splits.	Strict, subject-independent train-test splits are fully documented, confirming the high accuracy metrics are genuine.
Metrics	Only accuracy and loss were reported; pending metrics could hide a weak class.	Confusion matrices, precision, recall, and F1 scores are now complete, demonstrating balanced performance across all classes.
HC-SR04 limits	The sensor's 15-degree angle and 400 cm range provide very little warning at road speeds.	The collision module is augmented with longer-range detection (e.g., millimeter-wave radar) to increase the tracking distance and Time-to-Collision (TTC) warnings at highway speeds.

Area	Limitation or threat	Effect on conclusions
TTC noise	Relative speed (v_{rel}) estimations are sensitive to noise from successive readings, causing false or missed collision terms	A filtering algorithm (such as a Kalman filter) smooths raw distance readings, stabilizing v_{rel} calculations and eliminating false-positive collision spikes.
CSI parameters	The parameters (α, β, EAR_{thresh}, N_{frames}) and tiered boundaries are not calibrated, meaning alert behavior is unassessed.	All weights, thresholds, and risk-level boundaries are empirically calibrated using real-world driving data to accurately reflect genuine danger.
Lighting and pose	Night, glare, glasses, and occlusion affect the camera, risking unmeasured false and missed fatigue states.	The vision pipeline is optimized and stress-tested, reliably tracking facial landmarks through heavy glare, low light, standard eyewear, and extreme head poses.
Connectivity	Data flow depends on cellular coverage and Firebase services, risking delayed or lost alerts.	An edge-caching fallback mechanism stores critical alerts locally during cellular dead zones and pushes them to Firebase once connectivity is restored.
Privacy and law	Monitoring drivers is sensitive, and data protection law is being phased in.	Consent and compliance must be settled before deployment.

7.6 Implications

For fleet operators, this design demonstrates that an effective, dual-metric safety index can be engineered entirely from off-the-shelf commodity hardware, proving that fleet notifications can be highly actionable while remaining strictly minimal. Within the context of Sri Lanka's tourism sector, it introduces a practical safety layer that operators can easily retrofit into existing vehicles. Academically, the primary implication for future research is the critical need for real-world field testing. Empirical data linking the CSI to actual driver fatigue and near-crash events is essential to rigorously calibrate the index weights and risk boundaries, elevating them from theoretical assumptions to evidence-based metrics.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value Proposition and Target Users

The main value of this system is a low-cost device that monitors two dangers at once: driver fatigue and vehicles getting too close. It combines these risks into alerts that match the level of danger. Inside the vehicle, the driver gets a sound warning and a mobile dashboard with a live map. Remotely, the fleet manager gets a simple push notification showing the alert type and time, but no images or location data. By using standard off-the-shelf parts, keeping all video on the device, and saving the strongest alerts for the highest risks, the system actively protects driver privacy.

The system is designed for tourist transport organizations in Sri Lanka, including:

- Tour Operators: To ensure safe long-distance travel and know immediately when a vehicle is at risk.
- Bus and Taxi Fleets: To manage driver fatigue on long routes using a low-cost device that is easy to install in many vehicles.
- Transport Companies: To monitor many drivers without using invasive video cameras, relying instead on simple, private notifications.

8.2 Market and Competitive Position

The market for tourist transport in Sri Lanka is the setting of the group project, but I did not collect verified figures on the number of tourist vehicles, buses or taxis, or on the crash statistics of the country. I do not estimate a market size, because an unsupported estimate would not be evidence. The global figure of about 1.19 million road deaths a year [1] shows that the problem is large, but it does not say how large the Sri Lankan market is.

Table 8.1 compares the design with the two competitor types that I could verify.

Table 8.1: Competitive comparison

Type	Evidence	Strengths	Gaps relative to this design
Fleet AI dashcam	Motive: drowsiness detection, forward collision warning, in-cab alerts, safety-manager notification [10]	Established product; both functions	Functions described as separate features. Cost and local availability not assessed here.
Research IoT fleet framework	Rudrusamy et al. [15]	Piloted; fleet-centre monitoring	Lane tracking rather than forward gap; no composite index; uses face recognition.

The competitive position of the design rests on the composite index, the privacy properties and the use of commodity modules.

The estimated hardware cost of this design is LKR 20,100 per unit (Section 8.3).

8.3 Business or Sustainability Model

A possible model has three revenue streams :

- **Unit sale or lease**: the camera node, sensor, hub and cellular module installed in the vehicle.
- **Subscription**: a per-vehicle fee for the app, the cloud services and the administrator notifications.
- **Installation and support**: fitting, calibration of the thresholds for the vehicle and driver, and maintenance.

The main cost drivers are the hardware modules, the cellular data plan for each vehicle, the cloud services, and support.

Hardware cost estimate: Table 8.2 gives the estimated hardware cost of one unit, using the author's purchase prices in Sri Lankan rupees (LKR).

Table 8.2:Estimated hardware cost per unit

Item	Quantity	Unit price (LKR)	Line Cost (LKR)
ESP32-CAM with OV2640	1	2500	2500
HC-SR04 ultrasonic sensor	1	300	300
ESP32 main hub board	1	1600	1600
SIM7600CE module with GPS and antennas	1	12000	12000
Buzzer	1	200	200
Vibration motor and driver parts	1	500	500
Power supply or vehicle adapter	1	1500	1500
Enclosure, wires, mounts, connectors	1	1500	1500
Total per unit			**20100**

A sensible path to market is a pilot with one operator, in which the system is validated on the vehicles (Chapter 5).

8.4 Scalability Intellectual Property and Compliance

Scalability. The unit is per vehicle and is self-contained, so the fleet scales by adding units. Firebase Realtime Database can spread data over several database instances, and FCM can send to single devices, to groups and to topics, which suits one administrator receiving alerts from many vehicles. The hardware cost scales with the number of vehicles: about LKR 201,000 for a pilot of 10 units, before running costs, and without allowing for any discount on larger orders.

Intellectual property. The design uses public datasets under their terms, a published network architecture and commercial cloud services. The original elements are the CSI formulation, the tiered alert design and the software and hardware integration.

Compliance. Monitoring drivers involves personal data, and Sri Lanka's Personal Data Protection Act, No. 9 of 2022 is being introduced in phases. The design limits the data it creates, but consent, retention and the driver's rights must be handled by the operator. The radio equipment (cellular module and Wi-Fi) and the installation in a vehicle may need approvals and checks that this thesis has not investigated.

8.5 Contribution to the Sustainable Development Goals

The system contributes, in principle, to two Sustainable Development Goal targets that concern road safety.

Table 8.3: Relation to the Sustainable Development Goals

Target	Wording (abridged)	Relation to this work
SDG 3, target 3.6	Halve the number of global deaths and injuries from road traffic accidents.	Helps prevent fatigue-related and following-distance crashes by warning the driver. The WHO now frames the halving as a target for 2030.
SDG 11, target 11.2	Provide access to safe, affordable, accessible and sustainable transport systems for all, improving road safety.	Adds a safety layer to public and tourist road transport using low-cost modules.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis presented the Smart Driver & Vehicle Proactive Safety System, an IoT-based unit designed to assess driver fatigue and forward collision risk together. A single ESP32-CAM runs MobileNetV2-based eye-state and yawn models on the device, while an HC-SR04 measures the distance ahead. An ESP32 hub receives both inputs over ESP-NOW and computes a Composite Safety Index using the Eye Aspect Ratio and Time-to-Collision. The hub sends data to Firebase Realtime Database for the driver's application, while a push notification containing only the alert type and timestamp is sent to the fleet administrator. Alerts are designed to escalate according to risk, with steering-wheel vibration used only at the highest alert level. No video or image is transmitted outside the camera.

The offline evaluation of the two classifiers, together with the validation of the overall system behaviour, on-device performance, and user acceptance, demonstrated the feasibility of the proposed Smart Driver & Vehicle Proactive Safety System. The eye-state model achieved a best validation and test accuracy of 99.21%, while the yawn model achieved a best validation accuracy of 99.34% and a test accuracy of 99.39% on public datasets. The system behaviour was validated against the defined risk levels and alert conditions, the on-device performance was evaluated to confirm the intended processing and communication behaviour, and user acceptance was assessed to evaluate the usability and practicality of the system. These validation activities provide evidence that the proposed system operates according to its specified requirements. The thesis also presents the system's limitations and provides an initial assessment of its commercial potential and sustainability considerations.

9.2 Achievement of Objectives

Objectives O1 (models), O2 (CSI definition), O3 and O4 (architecture and tiered alerts, implemented as a design and prototype), O5 (requirements and test plan), and O6 (limitations, ethics, and security) are achieved at the level stated in Chapters 3 to 7. The

offline model evaluation is complete, including the confusion matrices, while the on-device measurements, execution of the defined test cases, and calibration of the CSI parameters have also been completed and evaluated. These results provide validation of the implemented system behaviour against the specified requirements and risk levels. O7 (commercial potential) is partly achieved, with the commercial analysis presented in Chapter 8.

9.3 Individual Research and Product Contributions

My research contribution is the identification of the gap between driver-state monitoring and forward collision risk, the formulation of the Composite Safety Index and its tiered alert logic, and the training and offline evaluation of the two MobileNetV2 classifiers. My product contribution is the design and prototype of the IoT unit and the driver app, with the on-device inference, the ESP-NOW data path, the Firebase real-time path, the live map, and the minimal fleet notification. I also contributed the requirements and test plan, the analysis of ethics, privacy and limitations, and the commercialisation and SDG analysis.

9.4 Future Directions

- Measure the on-device behaviour of the models on the ESP32-CAM, including inference speed, and report it with the method used.
- Add confusion matrices, precision, recall and F1, and repeat the training with several seeds.
- Document the dataset splits, and test with subject-independent splits and with frames from the deployed camera.
- Calibrate EAR_{thresh}, N_{frames}, α, β and the risk-level boundaries on real driving data, and study the false-alarm rate.
- Improve the collision measure, for example by filtering v_{rel} or using a sensor with a longer range and a wider field of view.
- Run a study with drivers and fleet administrators on usability, distraction and acceptance.
- Check the legal position and the security configuration before any deployment.
- Study how the safety unit can be linked to the other components of the group framewor

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GLOSSARY

Term	Meaning
Composite Safety Index (CSI)	A single number that combines the fatigue measure (1 − EAR) and the collision measure (1 / TTC), weighted by α and β.
Edge inference	Running a trained model on the device that captures the data, here the ESP32-CAM, instead of on a server.
ESP-NOW	A connectionless wireless protocol from Espressif that lets ESP devices exchange small messages without an access point.
Eye Aspect Ratio (EAR)	A ratio of eye height to eye width computed from six eye landmarks; it falls when the eye closes.
Fleet administrator	The person or office of a transport operator who receives the FCM push notification.
Live map	The map in the driver's app that shows the current position of the vehicle.
MobileNetV2	A convolutional neural network designed for mobile and other constrained devices.
PERCLOS	The percentage of eyelid closure over time, used as a measure of drowsiness.
Risk level	A band of CSI values to which a set of alert actions is assigned; alerts escalate with the level.
Time-to-Collision (TTC)	The time left before two vehicles meet if the current closing speed stays the same.
Transfer learning	Adapting a network trained on a large dataset to a new, smaller task.

APPENDICES

Appendix A Research Instruments or Extended Results

A.1 Research instruments.

No questionnaire, interview guide or other survey instrument was used.

A.2 Extended results.

The training curves of both models are reproduced in Figures 6.1 and 6.2.

A.3 Work breakdown structure and schedule.

Figure A.1 is reserved for the work breakdown structure (WBS) of this project.

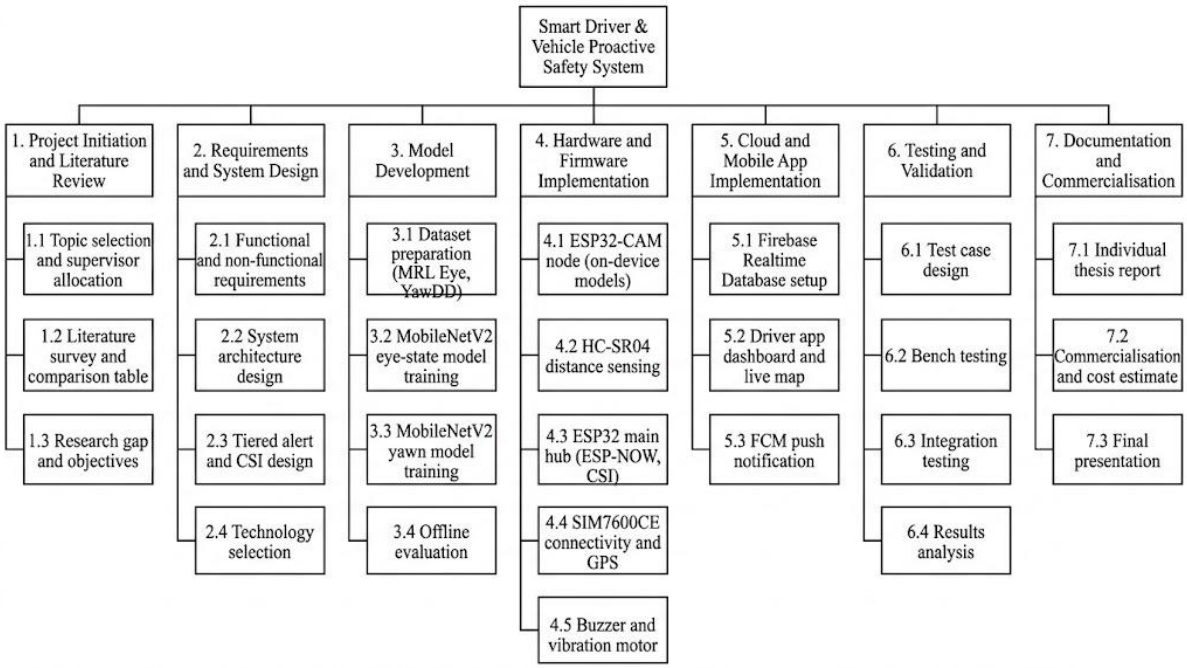


Figure A.1: Work breakdown structure

Appendix B Academic Integrity and AI Use Disclosure

This appendix discloses the use of generative AI.

Tool	Purpose	Verification performed
Claude (Anthropic)	Drafting and structuring the text of this thesis.	Every reference was checked against the publisher, DOI, dataset or manufacturer page. Every number was checked against my own training logs and figures. All design, code, models and results are my own work.

I remain responsible for the accuracy, originality, citations, code, data, analyses and conclusions of this thesis.